

CSCE 676 DATA MINING · MODULE 07 · LECTURE NOTES

Graph Mining

Structure in relationships.

MODULE OUTCOMES

- Explain the properties real networks have.
- Explain centrality measures and what each means.
- Explain PageRank and its interpretation.
- Explain community detection and modularity's limits.
- Compute these at scale.

1 What real graphs look like

- **Heavy-tailed degree distributions.** A few nodes have enormous degree while most have very little — which **breaks algorithms that assume bounded degree and dominates every average you compute**, so the mean degree is a nearly useless summary.
- **Short paths.** The **diameter grows like $\log n$ or slower**, so a breadth-first search reaches essentially everything within a few levels — **and the frontier explodes**, which makes BFS expensive in memory rather than in depth.
- **High clustering.** The **neighbours of a node are connected to one another far more often than chance would predict** — which is **what makes communities exist at all** (§3) and distinguishes real networks from random ones.
- **Sparsity.** The **average degree stays small even when n is enormous** — so an adjacency matrix is unthinkable at scale and an adjacency list is entirely manageable, which determines the representation.
- **And the heavy tail is the property that breaks things** — it causes **load imbalance in every partitioning scheme** (§4, and Module 10). **The practical consequence is immediate:** partitioning a graph by node gives one machine the hub and leaves the others with very little to do.

2 Centrality

Measure	What it means	Use it when
Degree	Many direct connections.	Immediate influence is what matters, and it costs $O(1)$ per node.

Measure	What it means	Use it when
Betweenness	Lies on many shortest paths between other nodes.	Flow and brokerage matter — and it is expensive (§4).
Closeness	Short paths to everybody else.	Reach matters; and it needs a connected graph, or a convention for disconnected pairs.
Eigenvector and PageRank	Connected to nodes that are themselves well connected.	Recursive importance is the right notion — and it scales (§4).

These measures disagree, and the disagreement is informative rather than a problem to be resolved — a node with high betweenness and low degree is a **bridge** connecting two otherwise separate regions, which is a genuinely different structural role from a **hub** with high degree and low betweenness. **That framing beats arguing about which measure is best**, which is not a well-posed question without saying what you want to know.

$$\mathbf{r} = d \mathbf{M}^T \mathbf{r} + ((1-d)/n) \cdot \mathbf{1}$$

- **A node's rank is the damped sum of the ranks flowing into it**, plus a uniform teleport term distributed over all nodes.
- **The interpretation is a random surfer's stationary distribution:** follow a random outgoing link with probability d , and jump to a uniformly random page with probability $1-d$ — which makes the quantity interpretable rather than merely computable.
- **It is computed by power iteration** — repeated multiplication by the transition matrix, which converges, and which is the sort of repeated local update that distributes well (§4).
- **And the teleport term is not a hack. It makes the Markov chain irreducible and aperiodic, which is exactly what guarantees a unique stationary distribution** that the iteration converges to — without it, dangling nodes and disconnected components break the whole thing. **Explaining it mathematically prevents it from looking arbitrary**, which it does in most presentations.

3 Communities

MODULARITY COMPARES EDGE DENSITY WITHIN GROUPS AGAINST A RANDOM BASELINE

It measures how many more edges fall inside the proposed communities than would be expected in a random graph with the same degree sequence — which is a reasonable definition of community structure and is cheap to compute for a given partition.

And maximising it is the standard approach — the Louvain and Leiden algorithms do this greedily and scale to graphs with billions of edges, which is why they are the practical default.

But it has a resolution limit: modularity maximisation cannot find communities below a size that depends on the *total* number of edges in the graph — so small but genuine communities get merged into larger ones, and the effect gets worse as the graph grows, which is a bad property for a method used at scale.

And it finds high-modularity partitions in random graphs — which is Module 01 §3 again: a high modularity score is not evidence that communities exist, without a comparison against a degree-preserving null model (§4's reporting requirement).

- **Louvain and Leiden** — greedy modularity optimisation, fast, and the practical default. **Leiden fixes a defect in Louvain by which the returned communities can be internally disconnected**, so prefer it.
- **Label propagation** — near-linear time, and **unstable across runs, which is itself informative** about whether the structure is real (Module 04 §3's stability argument).
- **Spectral methods** — principled, directly connected to graph cut objectives, and limited in practice by the cost of the eigenvector computation (Module 05 §2).
- **Overlapping community methods**, because **real community membership is not a partition** — people belong to several communities at once, and most algorithms return a partition anyway.
- **And assess the result by stability and by a null comparison**, not by the modularity score alone. **'Real membership is not a partition' is worth stating explicitly**, because **returning one is a modelling assumption rather than a finding** and is rarely flagged as such.

4 At scale

WHAT WORKS

anything expressible as repeated local updates:
 PageRank, label propagation, connected components,
 single-source shortest paths. Pregel's "think like
 a vertex" model fits these exactly, and so does
 sparse matrix multiplication.

WHAT DOES NOT

betweenness centrality -- it needs all-pairs
 shortest paths, so it is approximated by
 sampling source nodes instead
 exact triangle counting -- approximated, usually
 by sampling or by a sketch
 anything needing the full adjacency matrix

AND THE REAL PROBLEM IS PARTITIONING

cutting a heavy-tailed graph across machines puts
 the hub somewhere, and that machine then does most
 of the work. Edge-cut partitioning balances
 poorly; vertex-cut partitioning splits
 high-degree nodes across machines instead, which
 is what the real systems do.

SO: communication dominates, not computation.

Communication dominates rather than computation — which is why graph processing is substantially harder to distribute than most data-parallel work (CSCE 678 Module 11), where the partitions are independent. **The vertex-cut answer to the heavy tail is the key systems insight:** rather than trying to find a balanced edge cut in a graph where no balanced cut exists, split the problematic high-degree vertices themselves and reconcile their state.

AND WHAT TO REPORT ABOUT A GRAPH FINDING

How the graph was constructed — which edges you included is a modelling decision that determines every subsequent result, and it is frequently left unstated. A co-authorship graph with and without single-author papers, or a social graph with and without reciprocity required, are different objects.

Which centrality measure you used, and why that was the right question — since the four measures disagree (§2) and reporting 'the most central nodes' without saying which measure is uninterpretable.

The null comparison for any community finding — what a degree-preserving random graph scores on the same measure, which is the only way to read a modularity value at all (§3).

And the stability across runs and across subsamples, because several of these algorithms are non-deterministic and the variation is itself the result — a community structure that changes on every run has told you something, and it is not what the first run suggested.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Mining of Massive Datasets, chapters 5 and 10 (free)	§2's PageRank and §3's community detection , with the scale considerations throughout.
Easley & Kleinberg — Networks, Crowds, and Markets (free PDF)	§1 and §2 in their social context , free in full — and the clearest treatment of why these properties arise.
Fortunato & Barthélemy — Resolution limit in community detection (free)	§3's limitation, proved — short, and it should be read by anybody using modularity.
Malewicz et al. — Pregel (free)	§4's programming model — think-like-a-vertex, and why it fits this class of algorithm.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Plot the degree distribution** of a real graph on log-log axes.
- 2 **Compute the mean and median degree** and say why the mean misleads.
- 3 **Measure the diameter and the clustering coefficient**, and compare against a random graph with the same degrees.
- 4 **Compute all four centralities** and find a node that is a bridge rather than a hub.
- 5 **Implement PageRank by power iteration** and watch it converge.
- 6 **Set the teleport term to zero** and report what breaks.
- 7 **Run Louvain and Leiden** and compare the partitions.
- 8 **Run community detection on a degree-preserving random graph** and report the modularity it achieves.
- 9 **Run label propagation five times** and quantify the instability.
- 10 **Partition a heavy-tailed graph by edge cut** and measure the load imbalance.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name four properties of real graphs and the consequence of each.
- 2 Which property breaks things, and how?
- 3 Name four centrality measures and what each means.

- 4 Why is their disagreement informative?
- 5 Give the PageRank equation and its random-surfer reading.
- 6 Why is the teleport term necessary mathematically?
- 7 What does modularity measure, and what is its resolution limit?
- 8 Why is a high modularity score not evidence of communities?
- 9 What kinds of graph algorithm distribute well, and which do not?
- 10 Why does vertex-cut partitioning help?